

- 5 (a) (i) Absolute zero is the temperature at which all substances have a minimum internal energy.

Note: Candidates must relate absolute zero to temperature at which all substances have a minimum internal energy.

- (ii) The temperature of an ideal gas is directly proportional to the average kinetic energy of the molecules in the gas.

- (b) (i) Applying the equation of state of an ideal gas,

$$pV = nRT, \text{ amount of air}$$

$$n = 2.567 \text{ moles}$$

$$\text{mass of 2.567 moles} = 0.030 \times 2.567 = 0.077 \text{ kg}$$

- (ii) Applying the equation of state of an ideal gas,

$$pV = nRT, \text{ amount of air remaining}$$

$$n = \frac{pV}{RT} = \frac{(1.0 \times 10^5)(0.064)}{(8.31)(180 + 273)} = 1.700 \text{ moles}$$

$$\text{mass of air remaining} = 0.030 \times 1.700 = 0.051 \text{ kg}$$

$$\text{mass of air that escaped} = 0.077 - 0.051 = 0.026 \text{ kg}$$